

Michael Wu

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EDUCATION

California State Polytechnic University, Pomona

Expected May 2027

- Major: B.S. Computer Science
- Current GPA: 3.8 / 4.0, Dean's Honors List
- Coursework: Data Structures and Algorithms, Object Oriented Design and Programming, Artificial Intelligence, Software Engineering, Operating Systems, Computer Architecture, Formal Languages and Automata, Big Data Analysis and Cloud Computing, Prompt Engineering, Computer Organization and Assembly Programming, Unix and C programming

EXPERIENCE

Leidos QTC Health Services | Software Engineer Intern

May 2025 - Present

- Enabled 200+ contact-center agents to select outbound caller IDs by building a full-stack application for Amazon Connect.
- Reduced caller ID selection and dialing time by an estimated 40% by developing AWS Lambda functions that retrieved agent preferences from DynamoDB and initiated calls through the Amazon Connect CCP.
- Improved deployment consistency and reduced manual configuration by 80% using Infrastructure as Code for application resources.
- Leveraged Claude Code to identify and resolve 15+ code-quality and integration issues across React frontend and AWS backend.

QA Wolf | Software QA Engineer

July 2025 - May 2026

- Built 300+ end-to-end tests using JavaScript Playwright, streamlining regression cycles by 40% and reducing test escapes by 25%.
- Provided on-call support for clients, diagnosing critical test failures for businesses contributing over \$300k in revenue to QA Wolf.
- Balanced a full-time software engineering role with full-time studies, showcasing strong time management and problem-solving skills.

Northrop Grumman Collaboration Project | Software Engineer

June 2024 - May 2025

- Collaborated with 100+ engineers across Cal Poly Pomona and Cal Poly SLO campuses to research autonomous vehicle systems, ensuring successful end-to-end testing of 3 integrated subsystems.
- Engineered a telemetry simulation using C# and PostgreSQL, achieving 2.21ms latency and 10K+ packets/min across 3 data streams.
- Built scalable backend systems with Tokio and PostgreSQL, enabling sub-5 ms response times under 500+ concurrent requests.
- Designed a state management system using Rust for a multi-window Tauri PWA, reducing desynchronization across screens by 90%.

NASA Jet Propulsion Laboratory | Software Engineer (Open Source Contributor)

January 2025 - May 2025

- Contributing to OML Vision, a VSCode extension modernizing system modeling via web UI, enhancing usability for 100+ engineers.
- Diagnosed a macOS dependency and worked with JPL engineers to Dockerize development, cutting cross-platform setup time 60%.
- Implemented validated inline editing for OML data tables, reducing modeling errors 40% and accelerating complex dataset updates.
- Implemented notebook execution in OML Vision, enabling systems engineers to run external analysis tools (e.g., Python, R, MATLAB) from the GUI, reducing context-switching and boosting modeling efficiency by 30% in early user testing.

Computer Science Society ACM Chapter | President, Vice President, Project Initiative Chair

November 2023 - Present

- Lead CPP's ACM chapter, organizing workshops, industry partnerships, and professional development for 300+ students.
- Spearheaded a project initiative program empowering 100+ students to engage in hands-on computer science projects.
- Oversaw 15+ development teams coordinating timelines in Notion and standardizing version control practices using Git and GitHub.
- Launched a mentorship track pairing 30+ students with experienced developers, achieving 70% retention across project cycles.

PROJECTS

Lifeguard Vision | Next.js, FastAPI, MediaPipe, OpenCV, WebSockets, Git

- Built an AI web app for drowning detection, using real-time video and movement tracking, achieving 90% accuracy in test scenarios.
- Developed CV models to assess swimmer risk, using MediaPipe and OpenCV, boosting accuracy by 25% across diverse movements.
- Engineered a low-latency alert system using WebSockets, enabling real-time streaming and sub-100ms notifications at 99.9% uptime.

Icebreak | React Native, Node.js, Express.js, PostgreSQL, Prisma ORM, Amazon Web Services S3, Expo, Git

- Built a full-stack social media mobile app in a 20-person team, integrating 10+ core features over two development cycles.
- Designed guild pages in React Native, using Axios to display events, announcements, and members, improving user flow in demos.
- Developed RESTful API routes in Express.js, enabling full CRUD operations for guild events and attendance tracking, supporting simulated workflows for 5+ organization use cases.

SKILLS

Languages: Java, Python, JavaScript, TypeScript, C#, C/C++, Rust, SQL, Bash

Frameworks: React, React Native, Next.js, Node.js, Express.js, FastAPI, Tauri, Tokio, Playwright

Cloud & Tools: AWS (Lambda, API Gateway, DynamoDB, S3, Amazon Connect), Docker, Git, GitHub, GitLab, Postman

Databases & Other: PostgreSQL, MySQL, MongoDB, Redis, Prisma ORM, OpenCV, MediaPipe, WebSockets